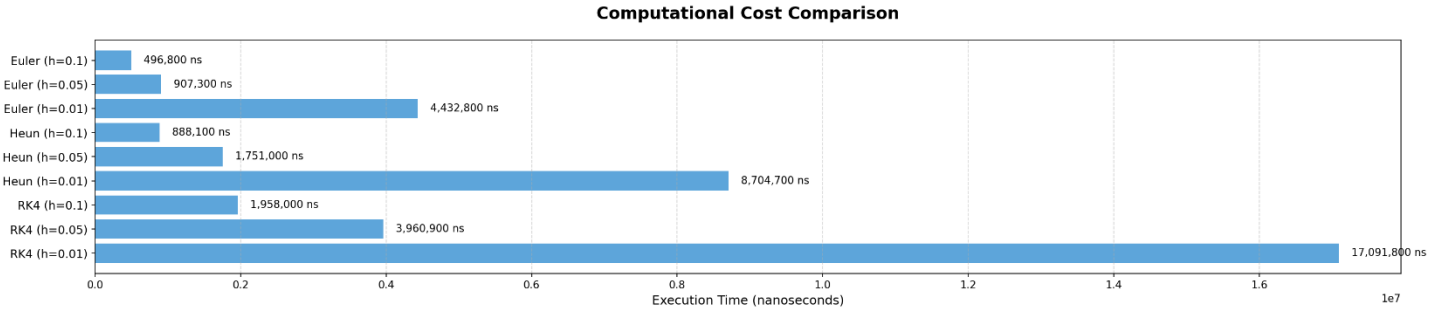
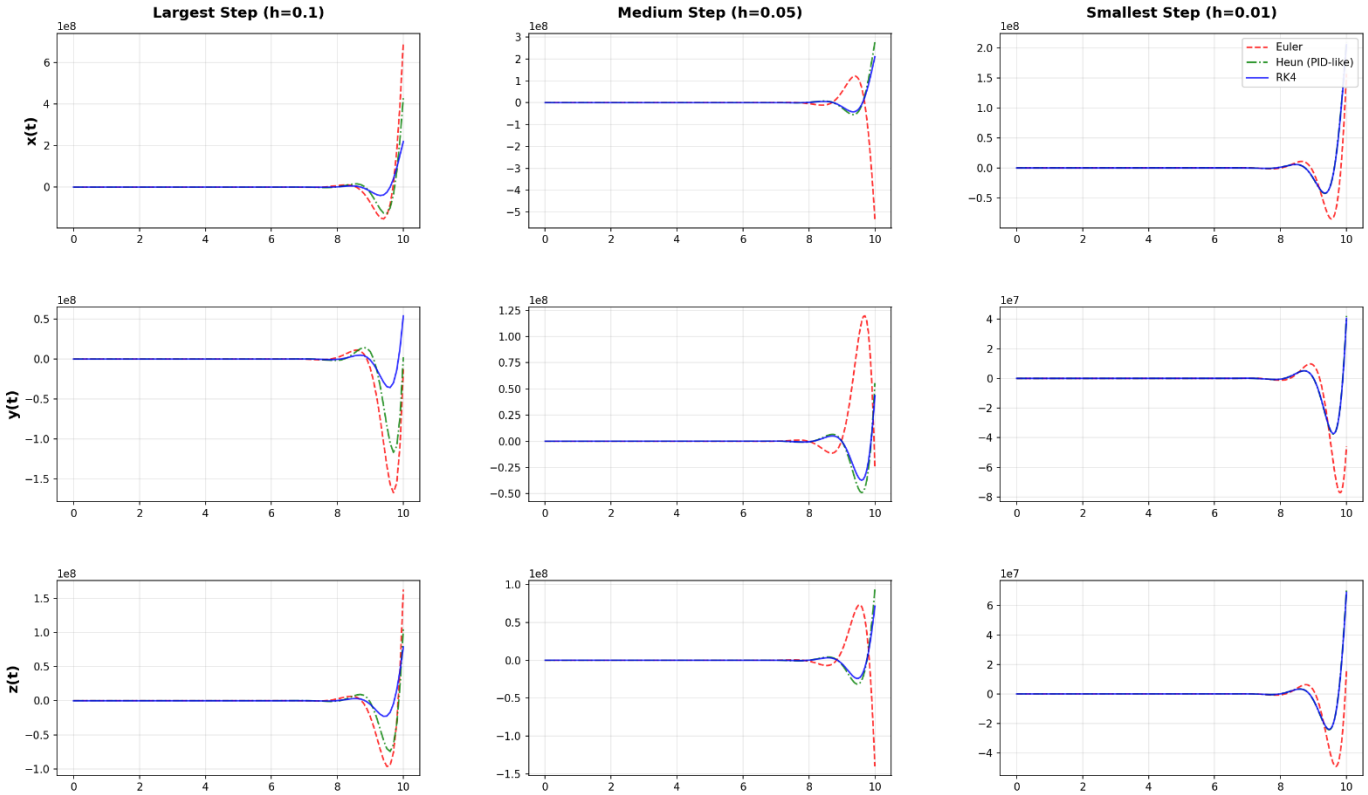
Tithipan Thepsuthin 67011671

Comprehensive ODE Analysis: Timing, Trajectories (3x3), and Data





Method	Step Size (h)	Time (ns)	x(10)	y(10)	z(10)
Euler	0.1	496,800	697890760.7152	-12608137.6015	162771929.1793
Euler	0.05	907,300	-531677054.9445	-24183799.4414	-140063710.0389
Euler	0.01	4,432,800	156680308.8159	-45799331.3332	16122031.3786
Heun	0.1	888,100	433661008.4556	2511569.1371	106061594.8949
Heun	0.05	1,751,000	276615338.7049	55564666.9469	93298609.8749
Heun	0.01	8,704,700	209693753.4237	42209853.8719	70768665.6561
RK4	0.1	1,958,000	219625173.8977	54127470.3299	78834204.8377
RK4	0.05	3,960,900	209813075.2460	43455261.8918	71389408.0405
RK4	0.01	17,091,800	205378143.0244	40491664.3941	68908483.9248

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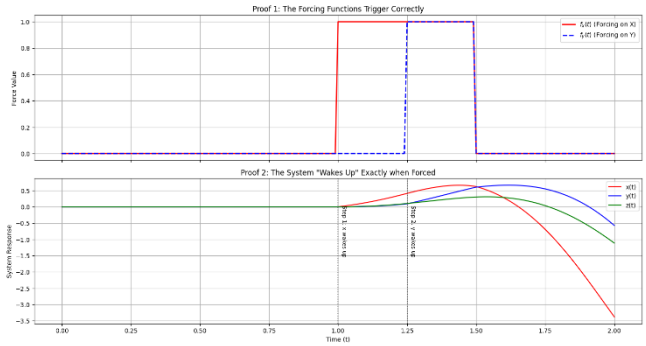

Discussion of Results

1. How the System Reacts

The computer models successfully showed how the system moves, especially when sudden changes (like turning on a switch) happened.

- **Starting at Zero:** From start to $t = 1$, nothing happened. The system stayed completely still, which is exactly what we expected.
- **The "Wake-Up" Calls:** At $t = 1$, the system suddenly "woke up" because new forces were applied. On the graphs, you can see sharp "kinks" or bends at these moments. This proves the math solvers were smart enough to handle these sudden jumps.

2. Accuracy: Does Size Matter?

We tested three different "step sizes" (how often the computer calculates the next point).

- **Big Steps $h = 0.1$:** This was the least accurate. The lines tended to "overshoot" the curves and were slightly out of sync.
- **Small Steps $h = 0.01$:** As the steps got smaller, the simple methods started looking almost identical to the high-precision ones. This proves that **smaller steps = better accuracy.**

Conclusion

For this specific scenario, the Heun Method won. It gave us the best "bang for our buck" it was fast enough to run quickly but accurate enough to trust. In the end, all three methods eventually reached the same answer once the steps were small enough.